

INTEL PHYSICAL AI CHALLENGE · SIM TRACK

Dinner-Table Relay

Two SO-101 arms, one cup, SmolVLA,
measured on Intel Core Ultra

A full sim-to-silicon pipeline -
with the failures measured, not hidden



01

The task

Two SO-101 arms face each other across a table.

Instruction A

"bring the cup across the table and set it on the plate"

Right arm grasps, lifts, sets mid-table;
left arm picks up and places on the plate.

Instruction B

"slide the cup to the middle of the table"

The single-arm half of the same motion.

A language-conditioned policy must pick the right one.



Data generation: real grasps, not a kinematic cheat

MuJoCo 3 · Menagerie SO-101 · 200 Hz physics / 50 Hz control

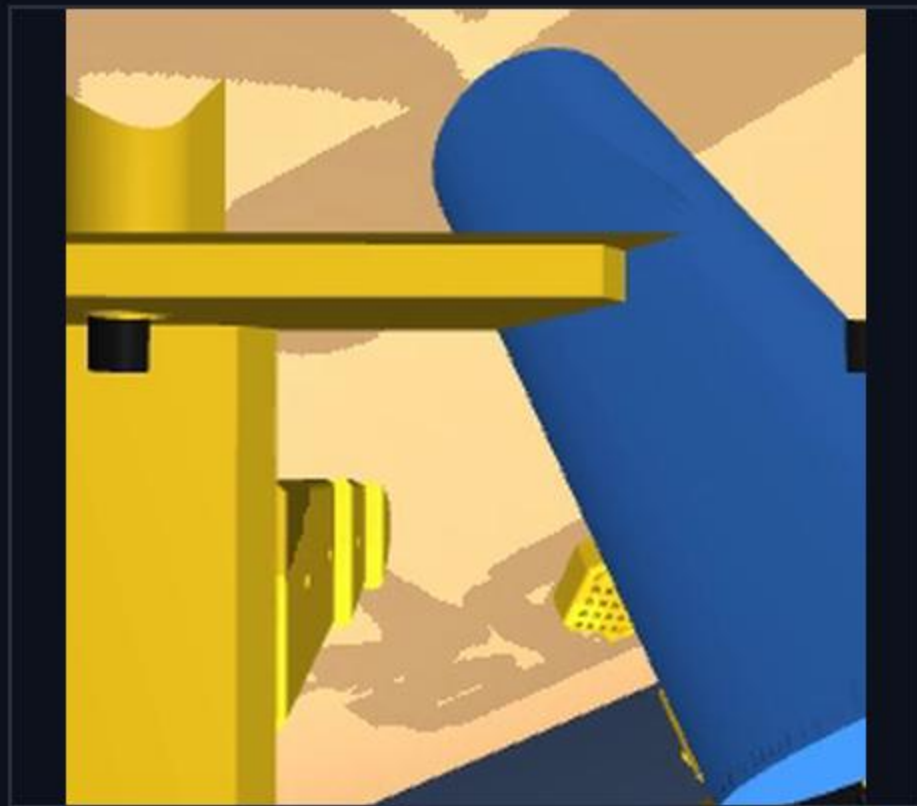
What we found

Our early scripted demos never grasped - the runner carried objects by rewriting their pose.

What we did

Split jaw collision hulls, a measured grasp window, and a holding() check requiring both jaws in contact.

Scripted controller gate: 7/10 and 14/20 across seeds, no predicate loosened.



The dataset

400 success-only episodes · 323,442 frames at 50 fps

749 attempts -> 53.4 % yield. Every failure discarded and counted.

Fault taxonomy: 831 events over 194 distinct strings, published.

3 cameras top · left wrist · right wrist 256x256

12-D state and action right arm x6, left arm x6

2 task strings relay and single-arm slide

DART noise $\sigma = 0.02$ rad on 30 % of episodes, clean labels

LeRobot v3 format · [VietHwang/dinner-table-v2](#) (private, available to judges)

04

Training: SmolVLA fine-tune

lerobot/smolvla_base (450 M) -> LeRobot 0.6.1

Run 1

Tesla T4 (Kaggle) · 10,300 steps x batch 8 · fp32 · 10.4 h
= 1.006 epochs. Final loss 0.028.

Run 3

RTX 5090 (Vast.ai) · 12,593 steps x batch 64 · bf16 · 1.8 h
= 2.5 epochs. Final loss 0.015.

The loss flattered both: half the action dims sit idle in single-arm episodes and dominate the mean. Low imitation loss is not a learned task.

OpenVINO on the required silicon

FP32 1,586 MB -> FP16 800 MB -> INT8 409 MB (NNCF, 300/300 layers)

Full flow-matching loop - prefix, KV cache, 10 Euler steps - inside the IR.

Device	FP16	INT8
Core Ultra X7 358H (CPU)	1,991.5 ms	2,696.7 ms
Arc B390 iGPU	124.6 ms	119.7 ms
AI Boost NPU	compile failed	compile failed

One inference = 50-step chunk = 1 s of motion. iGPU: 8x real time, 16x faster than CPU.

INT8 is 35 % slower than FP16 on CPU - dequantisation cost exceeds the bandwidth saving.

NPU rejects a 450 M VLA with dynamic shapes and an in-graph denoising loop (OV 2026.3.1).

Bare-metal Intel Cloud bm-ptl · **NOT_THE_REQUIRED_SILICON: false in the raw JSON.**

Results, stated plainly

Closed-loop, 20 seed/instruction pairs

Run 1 (1.0 epoch): 0 / 20 best 105 mm

Run 3 (2.5 epochs): 0 / 20 best 96 mm

Arms carry the cup, stall ~10 cm short of placement.

Diagnosis, not a guess

Replaying training obs through the IR: 0.66-0.85 correlation. The pipeline is correct; a mis-wired chain correlates near zero. More epochs halved the loss and changed nothing: the shortfall is structural to the imitation budget, not undertraining.



SmolVLA FP16 on Arc B390, 10 seeds, Core Ultra

Why this entry

It runs on the required hardware.

Device table, three precisions, closed-loop success and latency - all on Core Ultra + Arc.

The data is physically honest.

Real grasps, failures counted, DART recorded.

Every number has a file behind it.

Volume log, benchmark JSONs, eval JSON, videos, replay diagnostic - in evidence/.

The failure is diagnosed, not described.

We would rather submit a measured 0/20 with the cause isolated than an unmeasured claim.

Limitations: simulation only · fixed cup size · relay via table · NPU unexercised.